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Review of nuclear techniques for planetary science

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Spaceflight instruments that measure the energy and intensity of gamma rays and neutrons emitted by the surfaces of planetary bodies have been included in planetary science missions since 1966. These instruments use nuclear techniques to determine the elemental composition and related information about planetary surface and near-surface materials from the analysis of these gamma ray and neutron data. This paper provides a short review of these nuclear techniques as they have been used in current and past orbital and landed missions. A description of the nuclear instrumentation on many such missions is provided. Finally, descriptions of future missions that will employ gamma ray and neutron instruments will be presented.

Keywords: Planetary; nuclear; gamma; neutron.

1. Introduction

Measuring the energy and intensity of gamma rays and neutrons emitted by a planetary body can reveal a wealth of information about its near-surface bulk elemental composition. To date, orbiting and landed gamma-ray and neutron spectrometers have provided abundance measurements for H, C, O, Na, Mg, Al, Si, S, Cl, Ca, Ti, and Fe, as well as the naturally radioactive elements K, Th, and U from the Moon, Mars, Mercury and the large asteroids Ceres and Vesta.[1–9] Landed missions have measured the naturally radioactive elements on Venus in the past,[10] and currently a neutron instrument is being used to study the subsurface H content on the Mars "Curiosity" rover.[11] In the future, the composition of the all-metal asteroid 16 Psyche, the moons of Mars, and the surface of Saturn's moon Titan will all be studied using these gamma ray and neutron "nuclear" instruments. Knowing the chemical composition of a planetary surface is essential to understanding the processes that went into its formation and evolution.

This paper will discuss the nuclear processes involved in the production of gamma rays and neutrons by planetary materials, the instrumentation used to measure them, and the analysis methods employed to determine the bulk elemental composition of the emitting planetary body. The paper begins with a description of the nuclear processes by which Galactic Cosmic Rays (GCR) impinging on a planet result in the emission of gamma rays and neutrons and how the detection of this emission can be used to determine the bulk elemental composition of a planetary body.

Gamma ray and neutron spectrometers aboard orbital missions produce data used to create global maps of the distribution of elements over a planet's surface. Examples of relevant past and current orbital missions will be presented and their gamma ray and neutron instruments will be described.

The ability to land nuclear instrumentation directly on the surface of a planetary body presents new opportunities and challenges. These landed observations provide direct data to corroborate orbital measurements. Landed observations provide the ability to study the local surface composition at meter size scales rather than the 100s of kilometer scales typical of orbital observations. The "passive" measurements using GCR to produce gamma rays and neutrons are employed in both orbital and landed missions. However, some planetary bodies such as Venus, Titan and the Earth have atmospheres that are dense enough that the GCR particles are absorbed before reaching the planetary surface. In this situation, the missions must bring their own high energy neutrons to the surface. While a neutron-emitting radioactive source can be used, it is preferable to employ a Pulsed Neutron Generator (PNG) to stimulate the surface materials to produce gamma rays and neutrons. "Active" measurements employing a PNG in addition to gamma ray and neutron spectrometers will be addressed, and an example instrument currently operating on Mars will be discussed.

The paper concludes with a look to future planned orbital and landed planetary science missions that include such nuclear instrumentation and the scientific questions that will be addressed by their results.

2. Nuclear Processes Relevant to Planetary Science Instrumentation

Every body in the solar system is continuously bombarded by Galactic Cosmic Rays (GCR). GCR consist primarily of high energy (0.1–10 GeV) protons and alpha particles originating from outside the solar system. GCR have an intensity that varies slowly over the 11-year solar cycle, and hit a planetary body isotropically. GCR protons and alphas are sufficiently energetic that they typically penetrate about a meter into a planet's surface and interact with its material to produce a cascade of secondary particles. High energy (or fast) neutrons with energies up to ~20 MeV are one such particle with ~10 neutrons produced per incident high energy GCR proton. As shown in Fig. 1a, these high energy neutrons can interact with the nuclei of the planetary materials through inelastic neutron scattering to produce gamma rays with energies characteristic of the element involved. Thus, spectroscopic measurements of the gamma rays emitted by the planet can be used to determine its bulk elemental composition. Measurement of the gamma rays' energies identifies the elements present; measurement of their intensities provides the elemental abundances.

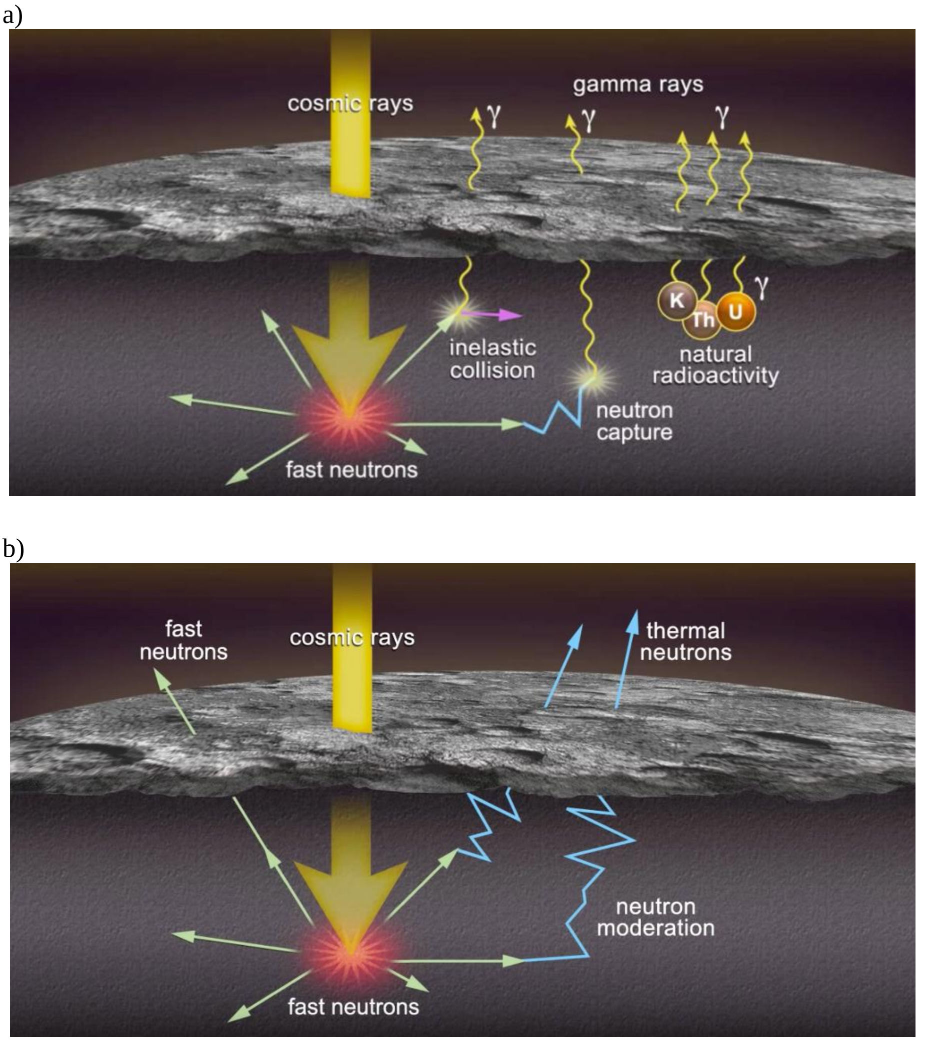

Fig. 1. These cartoons illustrate the different nuclear processes that result in the surface emission of a) characteristic gamma rays and b) fast, epithermal and thermal neutrons. (Image credit: NASA/The Johns Hopkins University Applied Physics Laboratory/Carnegie Institution of Washington).

The GCR-generated neutrons will lose energy as they scatter in the planetary material and will soon come to thermal equilibrium with their environment to become "thermal" neutrons. As shown in Fig. 1a, thermal neutrons can be absorbed by a nucleus through the process of thermal neutron capture, and as a result, gamma rays with energies characteristic of the element will also be emitted. As with the "inelastic" gamma rays resulting from the inelastic scattering process, these "capture" gamma rays can be used to identify the elements present and their concentrations.[1]

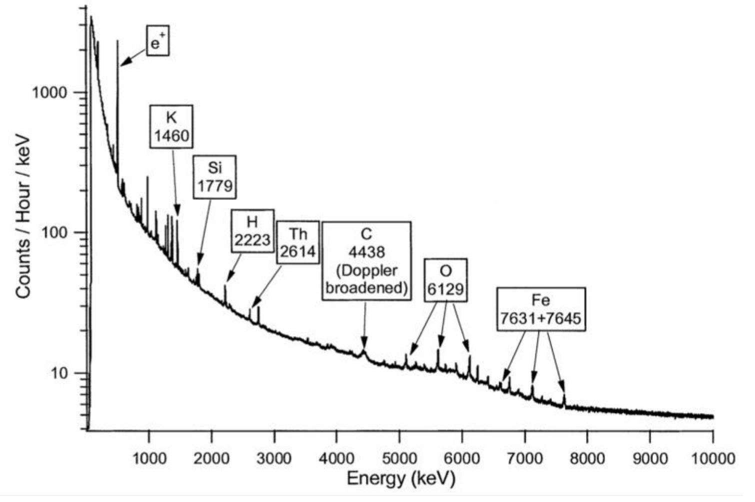

Fig. 2. A Mars Odyssey GRS gamma ray spectrum taken from June 10 through July 16, 2002. Several gamma ray emission lines are labeled with their energy in keV and the element responsible for the line.[5]

Gamma ray spectral data from the Mars Odyssey Gamma Ray Spectrometer (GRS)[5] are presented in Fig. 2. This histogram of gamma ray count rates as a function of energy shows spectral peaks at specific gamma ray energies for the elements labelled. Simple peak analysis can be used for data from spectrometers with high energy resolution such as high purity germanium (HPGe) to obtain a count rate associated with each element. These count rates can then be turned into relative, and sometimes absolute abundances of each element by weight (wt%).

Note that some neutrons exit (or leak out of) the surface of the planet and can be detected by external instrumentation, as was shown in Fig. 1b. These neutrons are categorized by their energies as "fast" ($E > 100$ keV), "epithermal" (0.4 eV $< E < 100$ keV) and "thermal" ($E < 0.4$ eV). When using neutron emission to map an area on the planetary surface, a relative suppression in the intensity of the epithermal neutrons is a strong indication of high hydrogen concentration at that location. This epithermal neutron suppression in the presence of high hydrogen content is a result of the fact that a neutron will lose the most energy if it collides with a proton (H nucleus) since their masses are essentially the same. This epithermal suppression effect is used extensively to identify regions of high H content on planetary bodies. One caution, however: there are other causes of epithermal neutron suppression that are not related to H concentration such as temperature and geochemistry variations.[12,13] Care must be taken in the interpretation of neutron data to account for these other possibilities.

Thermal neutrons have also been used to measure the abundance of neutron-absorbing elements iron, titanium, gadolinium, and samarium on the Moon[3,14,15] as well as CO_2 on Mars.[16] Fast neutrons carry information about iron and titanium (or, alternatively, average atomic mass) for nonhydrous silicate bodies.[17,18] Reference 19 presents particle transport simulations that establish that fast neutrons should dominantly vary with iron and titanium concentrations. They extended this analysis[18] to demonstrate that fast neutrons provide a robust measure of the average atomic mass of a planetary regolith. Effective atomic mass and bulk macroscopic neutron absorption cross sections can be used to characterize regional scale geological units, such as lunar mare and highlands.

3. Nuclear Instruments on Orbital Missions

There are many examples of orbital and landed robotic planetary science missions that include nuclear instrumentation such as gamma ray and neutron spectrometers. Table 1 provides summary information about the relevant orbital missions that will be described further in this section.

3.1. *Lunar Prospector*

The Lunar Prospector mission included both a gamma ray (GRS) and a neutron spectrometer (NS) to map the composition of the lunar surface. The GRS consisted of a bismuth germanate (BGO) scintillator inside a well-shaped anticoincidence shield (ACS) made of borated plastic (BC454) scintillator. The boron content of the BC454 allowed the ACS to also function as a neutron detector. The NS consisted of two ^{3}He filled gas proportional counters; one was covered with a thin thermal neutron-absorbing cadmium shield and so was only sensitive to epithermal neutrons. The second counter was covered with a thin Sn shield and so was sensitive to both thermal and epithermal neutrons. The

Table 1. Major current and past orbital missions with nuclear instrumentation aboard.

Mission	Planetary Body	Dates	Nuclear Instrumentation
Lunar Prospector	Moon	Launch: 1/1998 Mission End: 7/1999	GRS:BGO scintillator NS: ^{3}He tubes
Mars Odyssey	Mars	Launch: 4/2001; GRS: 2001-2009; NS, HEND 2001 -	GS: HPGe NS: Borated plastic scintillators HEND: ^{3}He tubes
MESSENGER	Mercury	Launch: 8/2004 Mercury Arrival: 3/2011 Mission End: 4/2015	GRS: HPGe NS: Borated plastic scintillator and two Li-glass scintillators
Dawn	Asteroids 1 Ceres and 4 Vesta	Launch: 9/2007 Mission End: 10/2018	GRaND: BGO, borated plastic and Li loaded scintillators
Lunar Reconnaissance Orbiter	Moon	Launch: 6/2009 Ongoing	LEND: collimated and uncollimated ^{3}He tubes

difference in counting rates of the two counters yields a measure of the flux of thermal neutrons. With its measurement of epithermal neutron suppression, the Lunar Prospector NS provided the first direct evidence of water-ice at the lunar poles.[3,20]

3.2. *Mars Odyssey*

The Mars Odyssey Gamma-Ray Spectrometer suite consisted of three different instruments: a passively-cooled high resolution HPGe gamma ray spectrometer (GS), a scintillator-based neutron spectrometer (NS), and a ^{3}He proportional counter-based high-energy neutron detector (HEND). These instruments worked together to collect the data needed for the mapping of elemental concentrations on the surface of Mars. Since the Mars Odyssey GRS carried no active anti-coincidence shield to reduce the field-of-view of the instrument, the GRS mapping "footprint" is comparable to the orbital altitude (~400 km). Over nearly 9 years, the Mars Odyssey GS produced a global map of the gamma ray emission over the entire midlatitude region between ±45˚ latitude.[21] The NS found hydrogen-rich deposits poleward of ±60°, which were interpreted as water-ice buried under a layer of desiccated soil.[22,23]

3.3. *MESSENGER*

The Mercury Surface, Space ENvironment, GEochemistry, and Ranging (MESSENGER) carried a mechanically cooled HPGe Gamma Ray Spectrometer (GRS) in Mercury orbit for ~15 months. The GRS instrument background was well characterized due to the use of its plastic scintillator particle shield and MESSENGER's highly eccentric orbit with periapsis at ~200–500 km altitude over the Northern hemisphere and apoapsis at ~15,000 km altitude in the South. Subtracting this instrument background allowed for greatly improved elemental abundance measurements.[7] The MESSENGER neutron spectrometer was composed of three scintillators, each optically coupled to a separate PMT in a "sandwich" configuration. The thermal neutron detector consisted of two GS20 Li-glass scintillators and the epithermal and fast neutron detectors BC454 borated plastic scintillator.[7]

3.4. *Dawn*

The Gamma Ray and Neutron Detector (GRaND) instrument aboard the Dawn mission visited the large asteroids 4 Vesta and 1 Ceres and mapped their surfaces using both gamma rays and neutrons. GRaND contained four types of radiation spectrometers: a large Bismuth germanate (BGO) scintillator, a Cadmium Zinc Telluride (CZT) semiconductor array, two boron-loaded plastic (BLP) scintillators and a phosphor sandwich (phoswich) of two more BLP scintillators and lithium loaded glass.[9] Dawn orbited Vesta for more than a year, from July 2011 to September 2012, and confirmed that Vesta is the parent body of the HED (howardites, eucrites, and diogenites) meteorites found on Earth. Dawn entered orbit around Ceres in March 2015 and discovered that it was an ocean world where water and ammonia reacted with silicate rocks. As the ocean froze, salts and other minerals concentrated into deposits that are now exposed in many locations across the surface.[24]

3.5. *Lunar Reconnaissance Orbiter*

The Lunar Reconnaissance Orbiter (LRO) was launched 18 June 2009 and entered its polar mapping orbit (~50 km circular) on 15 September 2009. LRO includes the Lunar Exploration Neutron Detector (LEND) instrument, containing collimated and uncollimated ^{3}He neutron spectrometers to map the lunar H content using the epithermal neutron suppression technique.[8]

4. Nuclear Instruments on Landed Missions

Instead of mapping the whole planet with pixels hundreds of kilometers wide, landed gamma ray and neutron instruments provide local bulk elemental composition data on meter scales and down to tens of cm depths. Such measurements provide corroborating measurements, or “ground truth” for orbital observations. These nuclear instruments make their measurements without coming in contact with the surface material, are rugged, and contain no moving parts, making them ideal for use on a rover or other relocatable vehicle.

Planetary bodies with dense atmospheres such as Venus, Titan and Earth do not permit GCR particles to reach the surface. In these cases, it is necessary for a landed nuclear instrument to also include a source of fast neutrons such as a deuterium–tritium (DT) PNG. A DT PNG emits 14.1 MeV neutrons essentially isotropically when it is turned on and emits no radiation when turned off. The ability to turn off the neutron source is an important distinction from neutron-emitting radioactive sources. This both eases safety concerns during mission integration on Earth and radiation damage concerns from spacecraft exposure to a radioactive source during the cruise to the planet to be studied. The 14.1 MeV energy of a DT PNG is high enough to permit the production of detectable high energy gamma rays from carbon, oxygen, and iron.

Even if the planetary body’s atmospheric density is low enough to permit passive measurements using GCR-generated neutrons, there are many advantages to using a PNG to make active measurements. The PNG average neutron intensity at 10^8 n/s is two orders of magnitude greater than the intensity of the neutrons produced by GCR interactions. As a result, using a PNG will allow elemental composition measurements to be made in 10% of the time needed for passive measurements for the same precision. DT PNGs emit their 14.1 MeV neutrons in pulses, allowing for the use of the time domain to reduce background and thus increase statistical precision in the measurements. Gamma rays produced during the PNG neutron pulses are primarily the result of inelastic neutron scattering. Between the pulses, the neutrons have enough time to thermalize so that thermal neutron capture gamma rays can be produced. Since the gamma ray energies are characteristic of both the emitting element and the specific nuclear process that created them, inelastic gamma ray spectra will have different characteristic gamma ray energies than the capture spectra even though the sample composition is the same. Thus, creating

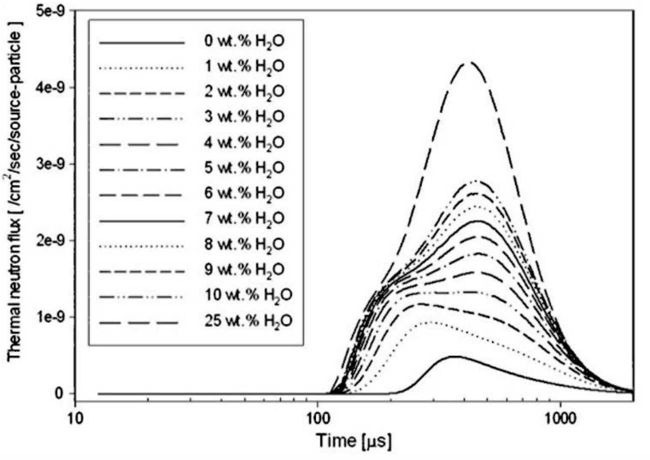

Fig. 3. Simulated arrival times at the detector for thermal neutrons (defined as energies less than 0.3 eV) for selected soil H_2O contents ranging from 0 to 25 wt%.[27]

separate spectra for the events that occur during the PNG neutron pulse and events that occur between the pulses will reduce the continuum background and greatly reduce problems of gamma ray peak interference leading to improved measurement precision.

The inclusion of a PNG on a landed mission also allows the use of the neutron die-away technique. Here, the detected neutron count rate is recorded as a function of time after the start of the PNG pulse. Figure 3 shows a simulation of thermal neutron die-away curves for different amounts of H in planetary materials. Note that the shape of the curve is strongly dependent on the amount of H present. The overall number of thermal and epithermal neutrons and the arrival times of thermal neutrons for a layered subsurface are a function of the hydrogen content of each layer, the layer depths, and the macroscopic absorption cross-section of each layer in the near-subsurface (~50 cm).[11,25,26]

4.1. *DAN on the Mars Science Laboratory ("Curiosity") Rover*

The Dynamic Albedo of Neutrons (DAN) instrument aboard the Mars "Curiosity" rover consists of two separate units: a DT Pulsed Neutron Generator (DAN/PNG) and the ^{3}He neutron Detectors and Electronics (DAN/DE) unit, which are installed at the two back corners of the rover. DAN uses the neutron die-away technique to provide estimates of, or constraints on, total subsurface hydrogen content, stratigraphy, and composition in the top ~50 cm of the Martian surface.[11]

5. Nuclear Instruments on Future Missions

In the future, both orbital and landed missions will carry gamma ray and neutron instruments to determine bulk elemental composition using nuclear techniques.

5.1. *Psyche*

NASA's Psyche mission will be launched in 2022 and arrive at the all-metal asteroid 16 Psyche in 2026. The Psyche mission will include a mechanically-cooled high-purity germanium (HPGe) gamma-ray spectrometer (GRS) surrounded by a borated plastic anticoincidence (AC) shield and a ^{3}He gas proportional counter-based neutron spectrometer (NS) to determine the composition of 16 Psyche over ~21 months. The mission's science goals are to understand planetary iron cores, a previously unexplored building block of planet formation, by examining an asteroid made of metal instead of rock and ice.[28]

5.2. *MMX/MEGANE*

The Japan Aerospace Exploration Agency (JAXA) has approved the Martian Moon eXploration (MMX) mission to be launched in 2024 to answer fundamental questions regarding Mars' two moons, Phobos and Deimos. Are these moons the result of captured asteroids[29] or were they formed via an impact of a larger body into Mars?[30] The lack of chemical composition information for these moons is a limiting factor in addressing these questions. MMX will use the Mars-moon Exploration with GAmma rays and NEutrons (MEGANE) instrument to make comprehensive remote sensing measurements of the composition of Phobos and Deimos, and then will return regolith samples of Phobos to Earth for further analysis. MEGANE includes a mechanically-cooled high-purity Ge (HPGe) gamma ray spectrometer surrounded by a borated-plastic anticoincidence shield (ACS). Thermal and lower-energy epithermal neutrons will be measured with two ^{3}He gas proportional counters.[31]

5.3. *Dragonfly/DraGNS*

NASA's Dragonfly mission, to be launched in 2026, will use a relocatable lander to study the pre-biotic surface chemistry of Saturn's moon Titan. This landed mission will use flight as a means of reaching many different locations on Titan to permit a broader understanding of the variety of surface chemistry processes found there. Dragonfly will include the Dragonfly Gamma ray Neutron Spectrometer (DraGNS) to measure the bulk elemental composition of the material directly beneath the lander to inform the decision of whether to take a sample for analysis by the Dragonfly Mass Spectrometer (DraMS) and to provide geochemical context for all of Dragonfly's measurements at each location. DraGNS will include a DT PNG, two gamma ray spectrometers (HPGe and $CeBr_3$ scintillator) as well as two ^{3}He neutron detectors. Dragonfly will arrive at Titan in early 2035 and spend nearly 3 years exploring its surface chemistry.[32]

6. Summary

Nuclear techniques provide valuable information toward the understanding of a planetary body's composition and origin. We have described and illustrated the nuclear processes by which neutron and gamma ray instruments can be used to measure the bulk elemental composition of a planetary body down to a few tens of cm below the surface. Both orbital and landed missions that include these nuclear instruments were described.

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